

A LONG-RANGE PROGRAM IN SPACE ASTRONOMY

Position Paper
of the
Astronomy Missions Board

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NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

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INTRODUCTION

The ability to mount astronomical instruments on observing platforms high above the Earth's atmosphere is the latest and most decisive in a series of technological developments that have stimulated the recent fantastic growth of astronomy. Astronomers are now able to observe the sky in ways they could only dream of less than 30 years ago. In those days, the known spectrum of radiation reaching the Earth's surface was only slightly broader in wavelength than the visible spectrum, 4000–7000 Å, because the Earth's atmosphere blocks out most of the shortwave radiation emitted by the Sun and the stars, and a good part of the infrared as well. Moreover, there was little reason to suspect that any significant number of astronomical bodies were radio emitters, and Jansky's early apparent observations of radio waves from the Milky Way aroused little interest among astronomers.

With observations restricted to a small fraction of the electromagnetic spectrum, a large part of astronomical research was concerned with the collection of data that could not be well understood. Astronomers could hope to deduce the physical nature only of objects emitting purely thermal radiation at a temperature of a few thousand degrees so that most of their radiated energy was radiated in the form of visible light. Very hot or very cold objects or those that gave off exotic, nonthermal radiation were beyond understanding. Then, one by one, the other regions of the spectrum began to be exposed. First the cosmic rays provided the first clue to high-energy processes in the universe. Next radio astronomy came to fruition, revealing both high-energy and nonthermal phenomena on a grand scale, notably the quasars and pulsars. Finally, the space program is bringing into view the ultraviolet region of the spectrum between 3000 and 300 Å, and X-rays and gamma rays at still shorter wavelengths.

No longer may theories be proposed that cannot be tested because key parts of the spectrum are inaccessible. On the contrary, it is now possible to devise and carry out decisive experiments to test almost any hypothesis in astrophysics, which is one of the reasons why more and more physicists now look upon astrophysics as the most interesting and exciting branch of physics.

Space astronomy offers two kinds of challenges. First, a great many well-defined scientific problems can now be solved by multi-

wavelength experimental approaches, and second, many unexpected discoveries are sure to be made as they always are when a new region of the spectrum is first explored or when a new instrument of unprecedented power is put into operation. The recent history of astronomy is full of examples of such unexpected discoveries. For example, the first radio and X-ray sources were both discovered accidentally and many of the recent discoveries of strong emitters of infrared radiation could not have been predicted in advance.

Telescopes in space have other important advantages beyond their capacity to intercept radiation that cannot penetrate the atmosphere. Thus, Earth-based telescopes must look through columns of turbulent air which severely degrade the images they produce. A telescope of 120-inch aperture above the Earth's atmosphere has 10 times the resolving power of the 200-inch telescope on Mount Palomar operating under the best atmospheric conditions. Because of the very small image sizes that are possible with telescopes in orbiting observatories, a space telescope 120 inches in diameter should be able to detect stars 100 times fainter than the faintest detectable from the Earth. Data on such faint objects are critical for settling major questions in cosmology, such as whether the universe is infinite or not.

To fully appreciate the unique contribution of space research in reaching otherwise inaccessible information about our universe, we should survey the great problems before us in modern astronomy, and determine for each case just what observations ought to be made next. This is how each of the reports of the subdisciplines (Part II) begins. Among the problems they examine are:

The quasars and the violent explosive events in the nuclei of galaxies which share many properties with those most distant objects known to man;

the strange problem of the million-degree temperatures in the solar corona surrounding the—astronomically speaking—cool surface of the Sun (a few thousand degrees) when, as every schoolboy knows, heat flows from hot to cold places and not the other way around;

the possibility that astronomers may be witnessing in some clouds of dust surrounding a certain star the formation of a system of planets like our own;

the mysterious pulsars, whose unnatural sounding rhythm led the first astronomers detecting them to catalog them as LGM 1, LGM 2, etc., where the LGM stood for “little green men”;

the puzzling situation in the interstellar medium where OH molecules (two-thirds of a water molecule), at temperatures hundreds of degrees below freezing (on Earth), are sending us brilliant maser beams of radiation, brighter than the radiation from any normal thermal source with a temperature of trillions of degrees; and

the most recent discovery of organic molecules existing in cold interstellar space—perhaps the simplest building blocks of life were not formed in early geological evolution but were created as part of the same process that formed the stars.

We conclude here with brief discussions of two challenging problems in modern astrophysics, and especially note the interplay between new techniques in widely separate subdisciplines.

One of these problems, the microwave background, is concerned with beginnings, with the cosmological question of the ultimate origin and fundamental forces that govern the evolution of the universe. The second example, the Crab Nebula, is related to endings, to the termination of the life of a star which apparently dies—contrary to T. S. Eliot's despairing poetic prediction—with a bang, not a whimper.

THE CRAB NEBULA: SUBJECT OF ALL DISCIPLINES

The Crab Nebula is now recognized as one of the most remarkable objects in the entire sky, combining the attentions of nearly every modern astronomical discipline, both spaceborne and ground based, observational and theoretical. The scribes of the Sung dynasty described it as a "guest star" that appeared on July 4, 1054, and in recent times the combined efforts of astronomers and scholars of Chinese history finally showed that the nebulosity known as M1 (object no. 1 in Messier's catalog of 1742) was indeed the remnant of the stellar explosion of 1054. The event was no ordinary nova (for "new" star) outburst, but an example of the much rarer supernova explosion, only three of which have been observed in our galaxy in the last thousand years. As a star exhausts its nuclear fuel, it cannot keep from collapsing under its own gravity. A complex series of events, whose details are still not completely understood, leads to a rapid collapse, followed by a violent explosion, during which the supernova releases more energy in 1 year— 10^{52} ergs—than it had given off in its entire lifetime as a star. Theorists have predicted for many years that the residue might include a neutron star—a star so compressed that atoms lose their individuality, their nuclei and electrons

merge, and the resulting state is best described as continuous nuclear matter. These neutron stars would be only 10 km or so in diameter, but would be extremely dense, with 1 cc weighing a billion tons or so. This corresponds approximately to the entire material of Manhattan Island, rock, buildings, and all, compressed into the volume of a thimble.

The combined efforts of radio, optical, and X-ray astronomers over the past 20 years have shown that the remnants of this incredible energetic explosion provide fascinating new ideas of the complexity of our universe. The optical astronomer sees a faintly glowing nebulosity, interlaced by a delicate network of red filaments that are expanding at a rate of 1000 km per second. In the center is a faint star, unlike any known stellar types. The light from the continuous part of the nebulosity was found to be strongly polarized, and by combining this information with the intense radiation seen by the radio astronomers, it became clear that the nebulosity is not just a glowing mass of hot gas, but a relativistic gas, composed of electrons whose energies exceeded 100 billion electron volts (and with velocities near the speed of light, the relativistic limit), an energy greater than that produced by any accelerator in operation on the Earth. The electrons emit radiation by the process known as synchrotron emission (like that given off by synchrotron accelerators), which requires that the relativistic gas be permeated by a magnetic field weak by terrestrial standards, but fabulously energetic when it extends over a region light-years in extent.

The advent of space technology brought forth the new field of X-ray astronomy, and after the initial discovery of Scorpio XR-1, the first X-ray star, the next object to be observed clearly as an X-ray emitter was the Crab Nebula. The X-rays might be generated by a "cool" gas at a temperature of a million degrees Kelvin—a substrate mingled with the relativistic gas—or by the relativistic electrons themselves. The evidence now favors the latter choice, but this implies many more energetic electrons, and at energies of 10 000 billion to a hundred thousand billion electron volts, which radiate so strongly that a new supply must be furnished continuously.

New surprises were in store for the astronomers, however, when an entirely new class of object—the pulsar—was discovered by radio astronomers two years ago. The first of these remarkable objects was observed to emit sharp radio pulses every second or so, with clocklike regularity. One of the suggested mechanisms that would explain the observations invoked the rotation of a

neutron star as the means of controlling the pulse rate, although the means of producing the radio pulses themselves remained unclear.

Within a year, a pulsar was discovered within the Crab Nebula, and in many respects the Crab Pulsar has proven to be the most promising key to the puzzle. Its repetition rate is 30 times per second—much faster than the typical pulsar—and the rapid rate can best be understood, it appears, by assuming that we are indeed observing a rapidly rotating neutron star. The optical astronomers quickly discovered that the peculiar star at the center of the Crab was indeed the pulsar, flashing in the visible as well as in the radio part of the spectrum. Space astronomy has now further extended the observations, for the X-ray astronomers have shown that approximately five percent of the X-rays from the Crab are pulsed in synchronism with the light and radio pulses.

The space observations have special significance because of the need for powerful energy sources to explain the source of X-ray emission. A hundred times more energy is emitted in the X-ray spectrum than in the visible, and the total rate of energy radiated by the Crab Pulsar must be over 100 times the rate of energy radiated by the Sun.

The combined X-ray, radio, and optical observations all support the model of the Crab Nebula being energized by a rotating neutron star. According to one theory, the rotation of the neutron star at 30 times per second results from the collapse of the original star, and the conservation of angular momentum. The collapse also compressed the normal magnetic field previously present in the star, to a value of a trillion gauss. Calculation then indicates that the huge electric field induced by the rapidly rotating magnetized star accelerates particles that drag the magnetic field with them, emitting synchrotron radiation as they go. In this way one accounts for the magnetic fields and relativistic particles observed in the nebula. Moreover, the process extracts energy and angular momentum from the star at a calculable rate, which agrees with the observation that the Crab Pulsar period is lengthening on a time scale of a thousand years.

Thus the discovery of a neutron star at the center of the Crab may solve several problems concerned with the energization of the nebula itself. More important, perhaps, it demonstrates the close relationship between the formation of neutron star and the explosion of a supernova. It will be of great interest to see whether all supernovae produce a pulsar (neutron star) to play a clocklike dirge during the final stages of evolution—the magnificent death

throes of a star, or whether some lead to a relativistic collapse to a gravitational singularity, a lump of matter so highly condensed that no radiation of any kind (and hence no information) can escape from its gravitational pull—"black holes" of the universe into which things may enter, but from which nothing ever returns.

THE COSMIC MICROWAVE BACKGROUND

The abundance of the chemical elements, and an unexplained discrepancy in the performance of a satellite communication system, unexpectedly prove to be closely related to the deep question of the origin and evolution of the universe. George Gamow and his coworkers showed theoretically 20 years ago that if the universe began as an initially compact mass of hot matter (the "big bang" theory of cosmology), they could explain some of the observed element abundances. A further consequence of the theory, not entirely appreciated at the time, was the lingering effect of the blast of gamma rays that would have been present in the initial fireball.

After the elements stopped forming, at a time when the universe would have been about 3 days old in the simplest cosmological model, the radiation began to degrade in energy, and should still be observable today. Gamow's colleagues estimated in 1949 that the cosmological red shift would have transformed the gamma rays in energy all the way through the electromagnetic spectrum to radiofrequencies, and the sky should today appear to have a uniform brightness of a "blackbody radiator" at a temperature of about five degrees Kelvin ($^{\circ}\text{K}$).

Four years ago two radio astronomers, Penzias and Wilson, showed that a previously discarded discrepancy in the observed noise from a very-high-sensitivity receiving system at the Bell Laboratories was indeed real and was observable no matter what direction in the sky they pointed their antenna. The apparent brightness of the sky at 7-centimeter (microwave) wavelength, when all corrections for the Earth, the atmosphere, and the galaxy had been applied, was measured to be about 3.5°K . Since then, measurements at many wavelengths between 3 mm and 70 cm have established that the spectrum is very accurately that of a blackbody radiating at a temperature of 2.7°K , and measurements in many directions confirm that it is a true isotopic cosmic background. Thus, the prediction of Gamow's theory was verified, and the range of acceptable cosmological models was greatly reduced. The steady-state theory, which envisioned a universe whose ap-

pearance never changes, with new galaxies constantly forming from new matter in the void left by the expanding system of old galaxies, cannot explain the new observation without additional hypotheses, such as a new class of radio sources far exceeding the number of galaxies. The radio evidence at present surely favors the big bang theory, but cosmological problems are notoriously slippery, and new observations are certainly needed to cross-check the new ideas.

If this cosmic microwave background radiation is a true "black-body" at 2.7° K, it should have its peak intensity at a wavelength of one millimeter and fall off rapidly at shorter wavelengths. Unfortunately, the Earth's atmosphere is opaque at this and shorter wavelengths, so it is difficult to verify the predicted decrease in intensity by ground-based measurements. Upper limits have been deduced at several wavelengths near one millimeter by observations of the state of excitation of interstellar molecules, and these suggest a fall in intensity, but the method is indirect. Several groups are therefore undertaking direct observations at millimeter wavelengths from balloons or rockets. One preliminary rocket observation had indicated that, to the contrary, there is a component of radiation at one millimeter which is almost 100 times stronger than a 2.7° K blackbody. If the observations of interstellar molecules are correct, the radiation observed by rockets must be locally produced.

The question of the ultimate origin and grand scheme of the universe is so important that one should be driven by only the utmost decisive evidence to acceptance of a particular cosmological model. The evidence for the "big bang" is mounting, but the final evidence is not in hand. Only through a combination of all the threads of observational evidence can the scientist be convinced. The choice of a particular theory for the origin of our universe has tremendous implications for many other astrophysical problems at the frontier of study, such as the nature and distribution of quasars, the origins, distribution, and dynamics of the high-energy cosmic rays, the abundance of the elements, and in particular the relative amounts of different isotopes. Those abundances must be determined more accurately in a variety of different astronomical objects. To be sure, our present best measurements of the helium abundance in the oldest stars support the cosmologies that are consistent with the cosmic microwave background, but the abundance determinations, in many instances, are still subject to large uncertainties.

Moreover, because of its great energy density, which exceeds

that of other known sources, the microwave background may have had profound effects on the evolution of galaxies. For example, density fluctuations which would otherwise have collapsed gravitationally to form galaxies may have been prevented from doing so by radiation pressure, and relativistic electrons ejected by distant quasars are rapidly decelerated by Compton collisions with microwave background photons. Examples such as these emphasize the importance of this new cosmological phenomenon.

The quest for the solution of the basic cosmological problem remains a difficult one, but the discovery of the cosmic microwave background has added new impetus and excitement. It permits us to look directly at radiation created when the universe was just a few days old, and allows us to surmise conditions in this most early history of the universe.